SEPARATION PROPERTIES DESCEND THROUGH PUSHOUTS ALONG CLOSED IMMERSIONS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19846-19853 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19855-19859, the complete author proof (3 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. BORDERLINE for Chief: the universally-closed claim on the pushout map is carried by the parent proposition rather than reproved inline; accepted as a named-input composition, flagged for possible pointer reclassification. Third pushout-series card (existence L16144, property transfer L16145).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter's pushout proposition (existence plus the universally closed coproduct map, card L16144's parent); Morphisms, lemma image-universally-closed-separated.

Source reference. Original lemma 09MQ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19859–19859. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E25: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19758–19761. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ECI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19708–19716. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0ECI. If X and Y are separated, then the pushout $Y \amalg_Z X$ (Proposition 0E25) is separated. Same with "separated over S", "quasi-separated", and "quasi-separated over S".*

Proof. The morphism $Y \amalg X \to Y \amalg_Z X$ is surjective and universally closed. Thus we may apply Morphisms, Lemma 09MQ. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.